Milestone 1: Social Returns of The Harbour, Victoria B.C.'s Supervised Consumption Site

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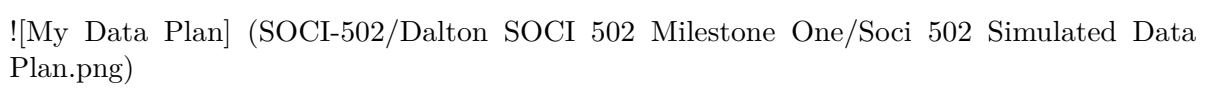
Victoria B.C.'s safe consumption site, The Harbour, is located on the city's infamous 900-block of Pandora Avenue. Operated by Island Health and SOLID Outreach, the facility is home to North America's first indoor group inhalation site and permits injection as well. The dataset used in this paper was collected as a part of a study funded by the British Columbia Ministry of Health and led by Dr. Michael Ma of the University of Victoria. This paper features simulated data for a select number of the dataset's response categories which have been choosen to answer this paper's research question.

Introduction: Description of Dataset and Research Question

As a student researcher on a study of The Harbour, Victoria's safe consumption site, I chose to use the survey data we collected as a part of this mixed methods study and further analyze it using R for my SOCI 502 Data Story project. Members (users) of The Harbour are housed or unhoused, use various types of drugs, range in age, race and gender, spend varying amounts of time at the site and for various purposes. The question I am asking is as follows: How is the time members spend at The Harbour associated with participation in non-consumption activities, and does this association vary with travel time to the site?

The research question will be considered answered by comparing the time participants spend at the site between those who report one or more non-consumption related reasons for visiting the site and those who report only consumption related reasons. Medians and percentage distributions will be used to compare these groups, and then again using short, medium and long travel time groups. If those who report non-consumption related activities have a consistently higher median, this will be considered a positive associated between non-consumption activities and time spent at the site. The same goes for travel time: if a difference between travel groups is notably larger or smaller in relationship to time spent for non-consumption related activities than consumption related ones, travel time will be shown to cause variation. In either case, if means are similar or the same, no association will be able to be interpreted.

Before I started on R and wrote my simulated code, I drew out my plan.



Loading Packages and Setting Up the Workspace

In order to simulate the data, the packages below need to be loaded. Given that they have previously been installed onto my machine, they now only need to be loaded.

```
library(tidyverse)
library(janitor)
library(extraDistr)
library(dplyr)
library(tinytable)
```

Simulating Variables

In order to answer the above question, I have chosen to use various response categories/survey questions from the dataset. The most important ones are "TimeAtHarb", "TimeToHarb", "CometoInject", "CometoSmoke," "CometoGetSubstances", "CometoDrugCheck", "CometoSeeFriends", "CometoSeeStaff", "ComeforODPrevention", "ComeforCleanSupplies", "ComeforSafety", "ComeforPrivacy", "ComeforWarmthShelter", "CometoChargePhone", "CometoBathroom", "ComeforCounselling", "ComeforDealers", "ComeforStreetNurse", "ComeforOtherServices".

The "TimeAtHarb" variable used response categories ranging from 1-8, for "TimeToHarb", responses varied (save some outliers) from 0-60 minutes, and each response to "Why do you come to The Harbour?" are in binary (1 = reason for coming, 0 = not a reason). These measures are taken into account when simulating the variables below. I have chosen to leave the outliers out of the data simulation as it's purpose is to ensure the code works rather than provide accurate data.

Generating the Data Frame

```r
library(dplyr)

set.seed(1836)
n_records <- 212

# Defining the 17 binary variable names.
var_names <- c(
  "CometoInject", "CometoSmoke", "CometoGetSubstances", "CometoDrugCheck",
  "CometoSeeFriends", "CometoSeeStaff", "ComeforODPrevention", "ComeforCleanSupplies",
  "ComeforSafety", "ComeforPrivacy", "ComeforWarmthShelter", "CometoChargePhone",
  "CometoBathroom", "ComeforCounselling", "ComeforDealers", "ComeforStreetNurse",
  "ComeforOtherServices"
)

# Defining probabilities for each variable (all set to 0.50).
probabilities <- rep(0.50, length(var_names))

# Simulating the matrix for the binary variables.
sim_matrix <- matrix(
  rbinom(n_records * length(var_names), size = 1, prob = probabilities),
  nrow = n_records,
  ncol = length(var_names),
  byrow = TRUE
)

# Converting to a data frame and adding column names.
visit_data <- as.data.frame(sim_matrix)
colnames(visit_data) <- var_names


# Now, adding the other two variables and adding the matrix to them in a tibble.
simulated_Harbour_data <- tibble(
  "Participant" = 1:n_records,
  "TimeAtHarb" = sample(
    x = c(1:8),
    size = n_records,
    replace = TRUE),
  "TimeToHarb" = sample(
    x = c(0:60),
    size = n_records,
    replace = TRUE)
```

```
) |>
  bind_cols(visit_data) # This attaches all 17 binary columns at once.

# Viewing the resulting simulated data frame.
#| tbl-pos: H
simulated_Harbour_data |>
  head(10)
```

```
# A tibble: 10 x 20
   Participant TimeAtHarb TimeToHarb CometoInject CometoSmoke
         <int>      <int>      <int>        <int>       <int>
 1           1          7         26            0           0
 2           2          2         13            0           0
 3           3          1         29            0           1
 4           4          6         30            1           0
 5           5          1         10            0           1
 6           6          3         14            0           0
 7           7          5         20            0           1
 8           8          1         12            1           0
 9           9          3         47            1           1
10          10          4         12            1           1
# i 15 more variables: CometoGetSubstances <int>, CometoDrugCheck <int>,
#   CometoSeeFriends <int>, CometoSeeStaff <int>, ComeforODPrevention <int>,
#   ComeforCleanSupplies <int>, ComeforSafety <int>, ComeforPrivacy <int>,
#   ComeforWarmthShelter <int>, CometoChargePhone <int>, CometoBathroom <int>,
#   ComeforCounselling <int>, ComeforDealers <int>, ComeforStreetNurse <int>,
#   ComeforOtherServices <int>
```

Now, I am making it into a table.

```
simulated_Harbour_data |>
  select(TimeAtHarb, TimeToHarb, 4:8) |>
  head(5) |>
  t() |>
  as.data.frame() |>
  tibble::rownames_to_column("Variable") |>
  tt() |>
  setNames(c("Variable", paste("Participant", 1:5)))
```

Variable	Participant 1	Participant 2	Participant 3	Participant 4	Participant 5
TimeAtHarb	7	2	1	6	1
TimeToHarb	26	13	29	30	10
CometoInject	0	0	0	1	0
CometoSmoke	0	0	1	0	1
CometoGetSubstances	1	1	0	0	1
CometoDrugCheck	0	0	0	0	1
CometoSeeFriends	0	0	1	1	0

Alternatively, this gives the sums of the reason columns.

```
tibble(Reason = var_names,
       n = colSums(simulated_Harbour_data[, var_names])) |>
  tt()
```

Reason	n
CometoInject	106
CometoSmoke	100
CometoGetSubstances	107
CometoDrugCheck	93
CometoSeeFriends	111
CometoSeeStaff	80
ComeforODPrevention	114
ComeforCleanSupplies	106
ComeforSafety	101
ComeforPrivacy	109
ComeforWarmthShelter	96
CometoChargePhone	110
CometoBathroom	102
ComeforCounselling	90
ComeforDealers	101
ComeforStreetNurse	103
ComeforOtherServices	111

References

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